Probability and Statistics

Sample Space in Probability

In **Probability Theory**, the sample space is the set of all possible outcomes of a random experiment. An experiment is any process that gives a result, like tossing a coin or rolling a die. Each outcome in the sample space is assigned a probability. A group of one or more outcomes from the sample space is called an **event**.

Sample-Space-of-Rolling-a-Die-and-Tossing-a-Coin

Another example of a Sample space for some random experiment:

- Experiment: Spinning a European roulette wheel
- Sample space: {0, 1, 2, 3, 4, 5, ..., 36}

Types of Sample Spaces

There are three types of Sample space:

Finite: The number of outcomes is countable and limited (e.g., rolling a die).
Sample Space: **S = {1, 2, 3, 4, 5, 6}**

Infinite: The outcomes go on forever (e.g., the Number of coin tosses needed until the first head appears).
Sample Space: **S = {1, 2, 3, 4, 5, 6, ... }**

Continuous: The outcomes can take any real value within a given range (e.g., A real number between 0 and 1).
Sample space : **S = {x ∈ R | 0 ≤ x ≤ 1}**

Finding Sample Space in Probability

To find the sample space in Probability, follow the steps below:

1. Identify all possible outcomes of the experiment.
2. List these outcomes in a set, ensuring each one is unique.

Combining sample spaces when multiple events occur helps calculate complex probabilities.

Sample Space- Tossing Coins

The sample space for tossing coins is given below:

When Two Coins are Tossed

Each coin can result in two possible events: either a head or a tail. In the case of flipping two coins, There are 4 outcomes in the sample space:
{HH, HT, TH, TT}

When Three Coins are Tossed

Sample Space for rolling 3 coins can be calculated keeping in mind the following:

When flipping three coins, the sample space encompasses all the possible combinations of heads and tails for the three coins.

It can contain **$2^3 = 8$** different outcomes, each with varying numbers of heads and tails in different orders.

Sample-Space-on-Tossing-Three-Coins

Sample Space- Rolling Dice

Sample space for rolling a die and two dices are given below:

When One Die is Rolled

On rolling a die, we can have 6 outcomes. So the sample space for rolling a die will be,
S = {1, 2, 3, 4, 5, 6}.

When Two Dice are Rolled

When rolling two dice, the sample space represents all the combinations of outcomes that can occur. It consists of pairs of numbers ranging from (1, 1) to (6, 6), consisting of **$6^2 = 36$** pairs. It helps in calculating probabilities for various sums or events involving two dice.

The following is a table of the sample space of rolling two dice.

Sample-Space-of-two-die

Solved Examples on Sample Space in Probability

Example 1: How many possible outcomes are there when rolling a fair six-sided die?

Solution:
There are 6 possible outcomes when rolling a fair six-sided die.

Example 2: In a deck of 52 playing cards, how many different ways can you draw two cards without replacement?

Solution:
There are 1,326 different ways to draw two cards from a deck of 52 playing cards without replacement.

Example 3: If you flip a coin three times, how many elements are in the sample space for this experiment?

Solution:
There are **$2^3 = 8$** elements in the sample space when flipping a coin three times.

Example 4: A jar contains 20 red marbles and 30 blue marbles. If you draw two marbles without replacement, how many different pairs can you get?

Solution:
A jar contains 20 red marbles and 30 blue marbles, so the total number of marbles is 50. When choosing two marbles, the pair can be Red–Red (RR), Blue–Blue (BB), or Red–Blue (RB), not only RB.

Therefore, the total number of ways to choose any pair of marbles from 50 marbles is:
{50 choose 2} = (50 × 49)/2 = 1225

Example 5: If you have a 4-digit PIN code, and each digit can be 0-9, how many possible PIN combinations are there?

Solution:
There are 10,000 possible PIN combinations for a 4-digit PIN code when each digit can be 0-9.

Practice Problems on Sample Space in Probability

Problem 1: If you flip a coin two times, how many elements are in the sample space for this experiment?

Problem 2: How many possible outcomes are there when rolling two fair six-sided dice simultaneously?

Problem 3: In a deck of 52 playing cards, how many different ways can you draw four cards without replacement?

Problem 4: In a deck of 52 playing cards, how many different ways can you draw two cards with replacement?

Problem 5: If you have a 3-digit PIN code, and each digit can be 0-9, how many possible PIN combinations are there?

Answers:

1. 4
2. 36
3. 270,725
4. 2,704
5. 1,000

Types of Events in Probability

An event is a particular outcome or a collection of outcomes from the sample space (the set of all possible outcomes) that we focus on or are interested in.

- The sample space (S) includes all possible outcomes.
- An event (E) includes one or more outcomes from that sample space that meet certain conditions.

Also, events can be classified into various types based on different properties and probability values of events.

It is now clear that events are subsets of the sample space. The following list gives the different types of events:

Impossible Event

These kinds of events are impossible and can be described by an empty set ∅ and are called Impossible Events.

Example:

Consider an experiment in which we roll a die. Now let's define an event that consists of outcomes that are multiple of 7. Sample space for this event is denoted by S,

S = {1, 2, 3, 4, 5, 6}

Now since there is no outcome in the sample space which is a multiple of 7. So, the set of event E will be an empty set.

Sure Event

Such an event, which has a probability of 1, i.e., occurrence of the event is certain or universal truth then that event is called Sure Event or Certain Event.

Note: An event that consists of the entire sample space is called a sure event (or certain event).

Example:

If we roll a die, as the event is the occurrence of a number less than 7, then it is sure that the occurring number is always less than 7 as the die only has numbers 1, 2, 3, 4, 5, and 6.

Simple Event

Any event that comprises a single result from the sample space is known as a simple event.

Example:

The Sample space of rolling a die, S = {1, 2, 3, 4, 5, 6} and the event for getting less than 2, E = {1}, where E has a single result taken from the sample space, Hence the event is a Simple event.

Compound Event

A Compound event is just opposite to what a simple event is, that is, any event that comprises more than a single result or more than a single point from the sample space, that event is known as a Compound event.

Example:

S = {1, 2, 3, 4, 5, 6} and E = {3, 4, 5}, where E is a Compound event.

Dependent Events

Dependent events are those in which the next outcome depends on the previous outcomes, which means, the probability of an event will change based on its previous outcomes.

Example:

Experiment: "Drawing balls from a bag consisting of 4 black and 3 red balls"

Event: "Drawing a black ball from the bag without replacement"

1. First draw.
   The probability of drawing a black ball = 4/7 = 0.571.

2. After the first draw, (if a black ball was drawn)
   Now the bag consist of 3 black and 3 red balls.
   Now the probability of getting black ball is 3/6 = 0.5

Thus, this event is dependent as the probability of each successive event depends upon the previous event.

Note: Here, there is a way of converting this dependent event into independent event, it can be done through Replacement. If after each experiment the ball is again kept in the bag, the sample space of the experiment will not change and hence, the probability of the event will remain same too.

Independent Event

Independent events are those in which the next outcome is independent of the previous outcome. This means the probability of the occurrence of an event will remain the same no matter how many times the same experiment is done.

Example:

Experiment: Rolling a die,
Event: "getting an even number"

1. Rolling the die : P(getting an even number) = 0.5,
2. Now the dice is rolled again, still P(getting an even number) = 0.5.

This means, that the probability of the event is independent of its previous outcomes.

Equally Likely Events

Those outcomes of an experiment that have the same probability are called Equally Likely Events. In other words, if two or more events have the same likelihood of happening, they are considered equally likely events.

Example:

Consider rolling a fair six-sided die. Each of the six possible outcomes (1, 2, 3, 4, 5, and 6) has the same probability of occurring, which is 1/6.

Therefore, rolling a 1 is equally likely as rolling a 2, 3, 4, 5, or 6.

Mutually Exclusive Events

Mutually exclusive events are events that cannot happen at the same time. If one happens, the other cannot.

Example:

When you roll a die:

- Event A: Getting a 2.
- Event B: Getting a 5.

These two events are mutually exclusive because you cannot get both 2 and 5 on the same roll.

Exhaustive Events

Exhaustive events are a set of events that cover all possible outcomes of an experiment. This means at least one of them must happen.

Example:

Tossing a coin:

- Event A: Getting Head.
- Event B: Getting Tail.

These two events are exhaustive because together they include all possible outcomes of the coin toss.

Problems on Types of Events

Problem 1: Two coins are tossed simultaneously. Define the events:
0. A : Getting at least one head

1. B : Getting no head.
2. Are A and B mutually exclusive?

Solution:
Sample space,
S = {HH, HT, TH, TT}

A = {HH, HT, TH}
B = {TT}

$A \cap B = \varnothing$

Since their intersection is an empty set, A and B are mutually exclusive events.

Problem 2: A die is rolled twice. Let event A be “sum of numbers obtained is 7” and event B be “first die shows 4.” Find A ∩ B.

Solution:
Sample space contains all ordered pairs (x, y) where x and y ∈ {1, 2, 3, 4, 5, 6}.

For event A:
A = {(1,6), (2,5), (3,4), (4,3), (5,2), (6,1)}

For event B:
B = {(4,1), (4,2), (4,3), (4,4), (4,5), (4,6)}

Now,
A ∩ B = {(4,3)}

Hence, A ∩ B = {(4,3)} — only one outcome satisfies both events.

Problem 3: A card is drawn from a standard deck of 52 cards. Let event A be “drawing a king” and event B be “drawing a red card.” Find A ∩ B and A ∪ B.

Solution:
Total cards = 52

Event A: King cards = {K♠, K♣, K♥, K♦}
Event B: Red cards = {All ♥ and ♦ = 26 cards}

A ∩ B = {K♥, K♦}
A ∪ B = {All red cards (26) + black kings (2)}

Hence, number of outcomes in A ∪ B = 26 + 2 = 28.

Problem 4: Two dice are rolled.
0. A: “Sum of numbers is even”

1. B: “Both numbers are even.”
2. Are these events mutually exclusive?

Solution:
Sample space has 36 outcomes.

For event A: sum even ⇒ (all outcomes where sum = even)
For event B: both numbers even ⇒ {(2,2), (2,4), (2,6), (4,2), (4,4), (4,6), (6,2), (6,4), (6,6)}

Now,
A ∩ B = {(2,2), (2,4), (2,6), (4,2), (4,4), (4,6), (6,2), (6,4), (6,6)}

Since intersection is not empty, A and B are not mutually exclusive events.

Probability Rules

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Probability rules are basic laws that help us calculate and understand the chances of events happening. These rules are used when solving probability problems in mathematics.

These fundamental principles, including the addition rule, multiplication rule, and complement rule, help determine the likelihood of events and calculate the probabilities of different outcomes in random experiments.

Various Probability Rules

Addition Rule

The addition rule of probability deals with the probability of the union of two events. If A and B are two events, then the probability of either event A or event B occurring is given by:

$P(A \cup B) = P(A) + P(B) - P(A \cap B)$

where,
$P(A \cap B)$ represents the probability of both events A and B occurring simultaneously.

This rule applies when events A and B are not mutually exclusive.

Example: Suppose we toss a fair coin. Let A be the event of getting a head, and let B be the event of getting a tail. The probability of either getting a head or a tail is given by:

$P(A \cup B) = P(A) + P(B)$
$= 1/2 + 1/2 = 1$

Multiplication Rule

The multiplication rule of probability applies when we want to find the probability of the intersection of two independent events. If A and B are independent events, then the probability of both events A and B occurring is given by:

$P(A \cap B) = P(A) \times P(B)$

This rule holds when the occurrence of one event does not affect the probability of the other event.

Example: Consider rolling a fair six-sided die. Let A be the event of rolling an even number, and let B be the event of rolling a number less than 4. Since these events are independent, the probability of rolling an even number and a number less than 4 is given by:

$P(A \cap B) = P(A) \times P(B)$
$P(A) = 3/6 = 1/2$
$P(B) = 3/6 = 1/2$
$P(A \cap B) = 1/2 \times 1/2 = 1/4$

Complement Rule

The complement Rule deals with the probability of the complement of an event. If A is an event, then the probability of the complement of A (i.e., the event not occurring) is given by:

$P(A') = 1 - P(A)$

Example: Suppose we draw a card from a standard deck of 52 playing cards. Let A be the event of drawing a heart. The probability of not drawing a heart (drawing a card that is not a heart) is given by:

$P(A) = 13/52 = 1/4$
$P(A') = 1 - P(A)$
$P(A') = 1 - 1/4 = 3/4$

Conditional Probability

Conditional probability is the probability of an event occurring given that another event has already occurred. If A and B are events, then the conditional probability of A given B is denoted by $P(A|B)$ and calculated as:

$P(A|B) = P(A \cap B) / P(B)$

This rule quantifies how the probability of one event changes in light of the occurrence of another event.

Example: Let's continue with the example of drawing a card from a standard deck of 52 playing cards. Suppose B is the event of drawing a face card (jack, queen, or king), and A is the event of drawing a heart. The probability of drawing a heart given that the card drawn is a face card is given by:

$P(A) = 13/52 = 1/4$
$P(B) = 12/52$
$P(A \cap B) = 3/52$
$P(A|B) = P(A \cap B)/P(B)$
$P(A|B) = (3/52)/(12/52) = 1/4$

This means that given the card drawn is a face card, there is a 1/4 chance that it is also a heart.

Sample Questions on Probability Rules

Question 1: You flip a fair coin twice. What is the probability of getting exactly one head?

Solution:
To solve this question, we can use the complement rule.

Probability of getting exactly one head is the complement of getting either two heads or two tails.

Probability of getting two heads: $P(HH) = (1/2) \times (1/2) = 1/4$
Probability of getting two tails: $P(TT) = (1/2) \times (1/2) = 1/4$

So, the probability of getting exactly one head is:
$1 - P(HH) - P(TT) = 1 - 1/4 - 1/4 = 1/2$

Question 2: A bag contains 5 red balls and 3 blue balls. You randomly select two balls from the bag without replacement. What is the probability of selecting one red ball and one blue ball?

Solution:
We can solve this question using the multiplication rule.

First, we find the probability of selecting a red ball, then the probability of selecting a blue ball after selecting a red ball.

$P(red) = 5/8$
$P(blue\ after\ red) = 3/7$

So, the probability of selecting one red ball and one blue ball is:
$P(red\ and\ blue) = P(red) \times P(blue\ after\ red) = (5/8) \times (3/7) = 15/56$

Question 3: You roll a fair six-sided die. What is the probability of rolling a number divisible by 2 or 3?

Solution:
We can solve this question using the addition rule.

We find the probability of rolling a number divisible by 2, the probability of rolling a number divisible by 3, and subtract the probability of rolling a number divisible by both 2 and 3 (i.e., 6).

Probability of rolling a number divisible by 2: P(divisible by 2) = 3/6 = 1/2
Probability of rolling a number divisible by 3: P(divisible by 3) = 2/6 = 1/3
Probability of rolling a number divisible by both 2 and 3: P(divisible by 6) = 1/6

So, the probability of rolling a number divisible by 2 or 3 is:
P(divisible by 2 or 3) = P(divisible by 2) + P(divisible by 3) - P(divisible by 6) = 1/2 + 1/3 - 1/6 = 5/6

Question 4: A card is drawn from a standard deck of 52 playing cards. What is the probability of drawing a face card or a heart?

Solution:
We can solve this question using the addition rule.

We find the probability of drawing a face card, the probability of drawing a heart, and subtract the probability of drawing a card that is both a face card and a heart.

Probability of drawing a face card: P(face card) = 12/52
Probability of drawing a heart: P(heart) = 13/52
Probability of drawing a card that is both a face card and a heart: P(face card and heart) = 3/52

So, the probability of drawing a face card or a heart is:
P(face card or heart) = P(face card) + P(heart) - P(face card and heart) = 12/52 + 13/52 - 3/52 = 22/52 = 11/26

Question 5: Two dice are rolled. What is the probability of getting a sum greater than 9?

Solution:
List outcomes: (4,6), (5,5), (5,6), (6,4), (6,5), (6,6) → 6 outcomes out of 36.
P(sum > 9) = 6/36 = 1/6

Practice Problems (with Answers)

1. A bag contains 5 red balls, 3 green balls, and 2 blue balls. What is the probability of drawing either a red ball or a green ball?
   Answer: 0.8

2. A fair coin is tossed twice. What is the probability of getting two heads in a row?
   Answer: 0.25

3. A deck of cards has 52 cards, 13 of which are hearts. What is the probability of not drawing a heart?
   Answer: 0.75

4. A standard deck of 52 playing cards is shuffled. If a card drawn is a face card, what is the probability that the card is also a spade?
   Answer: 0.25

Joint vs. Marginal vs. Conditional

Probability is a branch of mathematics that deals with the chance or likelihood of a random event occurring. In statistics, joint probability is a measure that tells us the likelihood of two events happening together at the same time, along with marginal and conditional probabilities.

Joint Probability

Joint probability is the probability of two (or more) events happening simultaneously. It is denoted as **P(A∩B)** for two events A and B, which reads as the probability of both A and B occurring.

For two events A and B, the joint probability is defined as:

$P(A \cap B)$ = P(both A and B occur)

Note: If A and B are dependent, the joint probability is calculated using conditional probability.

Example: Two fair dice are rolled at the same time. Find the probability that the first die shows 3 and the second die shows 5.

Solution:

The joint probability **$P(A \cap B)$** is the probability that the first die shows 3 and the second die shows 5.

Since the two dice are independent,

$P(A \cap B) = P(A) \times P(B)$.

Here, **$P(A) = 1/6$** and **$P(B) = 1/6$**.

So,

$P(A \cap B) = (1/6) \times (1/6) = 1/36$.

Marginal Probability

Marginal probability is the probability of an event occurring without considering other variables. It is found by adding up the joint probabilities of that event over all possible values of the other variables.

For two events A and B, the marginal probability of event A is defined as:

$P(A) = \sum_{B} P(A, B)$

Where **$P(A, B)$** is the joint probability of both events A and B occurring together. If the variables are continuous, the summation is replaced by integration:

$P(A) = \int_{B} P(A, B) , dB$

Example: Consider a table showing the joint probability distribution of two discrete random variables X and Y:

X/Y	Y = 1	Y = 2
X = 1	0.1	0.2
X = 2	0.3	0.4

To find the marginal probability of **X = 1**:

$P(X = 1) = P(X = 1, Y = 1) + P(X = 1, Y = 2) = 0.1 + 0.2 = 0.3$

Conditional Probability

Conditional probability is the probability of an event happening when another event has already occurred. It helps us adjust our prediction of an event using the information we already know.

The conditional probability of event A given event B is denoted as **$P(A|B)$** and is defined by the formula:

$P(A|B) = P(A \cap B) / P(B)$

Where:

- **P(A∩B)** is the joint probability of both events A and B occurring.
- **P(B)** is the probability of event B occurring.

Example: Suppose a card is drawn at random from a standard deck of 52 playing cards. Find the probability of drawing an Ace given that the card drawn is red.

Solution:

Let A be the event of drawing an Ace and B be the event of drawing a red card.

In a standard deck of 52 cards, there are 26 red cards. Out of these, 2 are red Aces (Ace of Hearts and Ace of Diamonds).

The required probability is:

P(A|B) = P(A ∩ B) / P(B)

Here,
P(A ∩ B) = 2/52 and **P(B) = 26/52**

So,

P(A|B) = (2/52) / (26/52) = 2/26 = 1/13.

Bayes Theorem in Machine learning

Bayes Theorem explains how to update the probability of a hypothesis when new evidence is observed. It combines prior knowledge with data to make better decisions under uncertainty and forms the basis of Bayesian inference in machine learning.

- Handles uncertainty and noisy data effectively
- Supports probabilistic interpretation of model predictions
- Enables learning even with limited data
- Plays a key role in real-world applications such as risk analysis and diagnosis

Mathematical Formulation of Bayes Theorem

Bayes Theorem describes the relationship between conditional probabilities and is mathematically expressed as:

P(A | B) = \frac{P(B | A) P(A)}{P(B)}

where

- **Posterior Probability P(A | B)**: the updated probability of hypothesis A after observing evidence B
- **Likelihood P(B | A)**: the probability of observing evidence B assuming hypothesis A is true.
- **Prior Probability P(A)**: the initial belief about hypothesis A before observing any evidence.
- **Evidence (Marginal Likelihood) P(B)**: the total probability of observing evidence B acting as a normalization factor.

Bayes Theorem for Multiple Hypotheses (n Events)

For a set of mutually exclusive and collectively exhaustive hypotheses { E_1, E_2, \dots, E_n } and an observation O the generalized Bayes' Theorem is given by:

P(E_i | O) = \frac{P(O | E_i) P(E_i)}{\sum_{j=1}^{n} P(O | E_j) P(E_j)}

where

- P(E_i | O) is the posterior probability of hypothesis E_i
- P(O | E_i) is the likelihood of observing O under hypothesis E_i
- P(E_i) is the prior probability of E_i

Step By Step Implementation

Here in this code we implements a **Naive Bayes classifier** that uses Bayes Theorem to compute the probability of a message being spam or ham based on word frequencies, trains the model on labeled data, evaluates its performance and predicts the class of new unseen messages.

Step 1: Install and Import Required Libraries

- Install essential Python libraries for data handling, ML modeling and visualization
- pandas is used for data manipulation
- scikit-learn provides ML algorithms and evaluation metrics
- matplotlib and seaborn are used for visualization

```
pip install pandas scikit-learn matplotlib seaborn
import pandas as pd
from sklearn.model_selection import train_test_split
from sklearn.feature_extraction.text import CountVectorizer
from sklearn.naive_bayes import MultinomialNB
from sklearn.metrics import accuracy_score, confusion_matrix, classification_report
import seaborn as sns
import matplotlib.pyplot as plt
```

Step 2: Load and Preprocess the Dataset

- Load the spam dataset using pandas
- Convert categorical labels into numeric values

You can download dataset from here

```
df = pd.read_csv("spam.csv", encoding='latin-1')[['v1', 'v2']]
df.columns = ['Label', 'Message']
df['Label'] = df['Label'].map({'ham': 0, 'spam': 1})
print(df.head())
```

Output: (shows a table with columns Label and Message – image reference: bayes1)

Step 3: Separate Features and Target Variable

- Text messages are taken as input features
- Labels (0 for Ham, 1 for Spam) are the target variable

```
X = df['Message']
y = df['Label']
```

Step 4: Split Data into Training and Testing Sets

- Split dataset into 80% training and 20% testing
- random_state ensures reproducibility

```
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
```

Step 5: Convert Text Data into Numerical Form

- Machine learning models cannot process raw text
- CountVectorizer converts text into word frequency vectors

```
vectorizer = CountVectorizer()
X_train_vec = vectorizer.fit_transform(X_train)
X_test_vec = vectorizer.transform(X_test)
```

Step 6: Train the Naive Bayes Model

- Multinomial Naive Bayes is well-suited for text classification
- Train the model using vectorized training data

```
nb_model = MultinomialNB()
nb_model.fit(X_train_vec, y_train)
```

Step 7: Make Predictions on Test Data

- Use the trained model to predict spam or ham messages
- Predictions are stored for evaluation

```
y_pred = nb_model.predict(X_test_vec)
```

Step 8: Evaluate Model Performance

- Calculate accuracy to measure overall performance
- Generate confusion matrix for detailed analysis
- Display precision, recall, and F1-score

```
accuracy = accuracy_score(y_test, y_pred)
print(f"Accuracy: {accuracy*100:.2f}%")
cm = confusion_matrix(y_test, y_pred)
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', xticklabels=['Ham', 'Spam'], yticklabels=['Ham', 'Spa
plt.title("Confusion Matrix")
plt.xlabel("Predicted")
plt.ylabel("Actual")
plt.show()
print(classification_report(y_test, y_pred))
```

Output: (Confusion Matrix heatmap – image reference: Bayes2)
Accuracy value and classification report are displayed.

Step 9: Test the Model on New Messages

- Use the trained model to classify unseen messages
- Vectorize new messages using the same vocabulary

```
new_messages = [
    "Congratulations! You won a free ticket.",
    "Hey, are we still meeting today?"
]
new_vec = vectorizer.transform(new_messages)
predictions = nb_model.predict(new_vec)
for msg, pred in zip(new_messages, predictions):
    label = "Spam" if pred == 1 else "Ham"
    print(f"Message: '{msg}' => Prediction: {label}")
```

Output:

Message: 'Congratulations! You won a free ticket.' => Prediction: Spam

Message: 'Hey, are we still meeting today?' => Prediction: Ham

You can download full code from here

Applications of Bayes Theorem in Machine Learning

Bayes Theorem ability to handle uncertainty and incorporate prior knowledge allows models to make accurate predictions even with incomplete or noisy data like:

1. **Naive Bayes Classifier**: It is a simple probabilistic model based on Bayes' theorem that assumes feature independence, making it efficient and effective for tasks like text classification and spam detection.

2. **Bayes optimal classifier**: The Bayes optimal classifier is a theoretical model that predicts the class with the highest posterior probability for given features, representing the best possible classification accuracy. It uses Bayes' theorem to update probabilities based on new evidence.

   \hat{y} = \arg \max_{y} P(y | x)

where

- \hat{y}: Predicted class label
- y: Possible class label
- x: Input feature vector
- P(y | x): Posterior probability of class given features

3. **Bayesian Optimization**: Bayesian Optimization is a technique for efficiently finding the maximum or minimum of expensive to evaluate functions using a probabilistic model often a Gaussian process. It iteratively selects the most promising points to evaluate, making it ideal for tasks like hyperparameter tuning in machine learning.

4. **Bayesian Belief Networks (BBNs)**: Bayesian Belief Networks (BBNs) or Bayesian networks are probabilistic graphical models that represent variables and their conditional dependencies using a directed acyclic graph (DAG). They are widely applied in risk analysis, diagnostics and decision-making.

Probability Distributions in Data Science

Understanding how data behaves is one of the first steps in data science. Before we dive into building models or running analysis, we need to understand how the values in our dataset are spread out and that's where probability distributions come in.

Example:
If you roll a fair die, the chance of getting a 6 is 1 out of 6, or 16.67%. This is a basic example of a probability distribution — a way to describe the likelihood of different outcomes.

Probability Data Distributions

When dealing with complex data like customer purchases, stock prices, or weather, probability distributions help answer:

- What is most likely to happen?
- What are rare or unusual outcomes?
- Are values close together or spread out?

This helps us make better predictions and understand uncertainty.

Why Are Probability Distributions Important?

- Explain how data behaves (clustered or spread)
- Form the basis of machine learning models
- Used in statistical tests (e.g., p-value)
- Help identify outliers and make predictions

Before this, we need to understand **random variables**, which assign numbers to outcomes of random events (e.g., rolling a die).

Key Components of Probability Distributions

Now that we understand random variables let's explore how we describe their probabilities using three key concepts:

1. **Probability Mass Function (PMF)**: Used for discrete variables (e.g., number of products bought). It gives the probability of each exact value. For example, 25% of customers buy exactly 3 products.

2. **Probability Density Function (PDF)**: Used for continuous variables (e.g., amount spent). It shows how probabilities spread over a range but not the chance of one exact value since values can be infinite.

3. **Cumulative Distribution Function (CDF)**: Used for both types, it shows the probability that a value is less than or equal to a certain number. For example, CDF(3) = 0.75 means 75% buy 3 or fewer products; CDF$(50) = 0.80 means 80$50 or less.

   To find the CDF we can use the formula given below:
   CDF: (F_X(x) = P(X \leq x) = \int_{-\infty}^x f(t) , dt)
   Where (F(x)) is the CDF and (f(t)) is the PDF.

Types of Probability Distributions

Probability distributions can be divided into two main types based on the nature of the random variables: **discrete** and **continuous**.

Discrete Data Distributions

A discrete distribution is used when the random variable can take on countable, specific values. For example, when predicting the number of products a customer buys in a single order the possible outcomes are whole numbers like 0, 1, 2, 3, etc. You can't buy 2.5 products so this is a discrete random variable.

1. Binomial Distribution

The binomial distribution calculates the chance of getting a certain number of successes in a fixed number of trials.

Formula (PMF):
(P(X = k) = \binom{n}{k} p^k (1-p)^{n-k})

Where:

- (n) = number of trials
- (k) = number of successes
- (p) = probability of success

Example: Flipping a coin 10 times and counting heads.

- Number of trials: 10
- Two outcomes per trial: heads (success) or tails (failure)
- Probability of success (heads): 0.5
- Shows likelihood of getting 0 to 10 heads

2. Bernoulli Distribution

The Bernoulli distribution describes experiments with only one trial and two possible outcomes: success or failure. It’s the simplest probability distribution.

Formula (PMF):
(P(X = x) = p^x (1-p)^{1-x}, \quad x \in {0,1})

Example: Flipping a coin once and checking if it lands on heads.

- One trial only
- Two outcomes: heads (success) or tails (failure)
- Probability of success: 0.5

3. Poisson Distribution

The Poisson distribution models the number of random events happening in a fixed time or area.

Formula (PMF):
$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}$$

Where:

- λ = average rate of occurrence

Example: Counting how many customers enter a coffee shop per hour. It helps predict the probability of seeing a specific number of events based on the average rate.

4. Geometric Distribution

The geometric distribution models the number of trials needed to get the first success in repeated independent attempts.

Formula (PMF):
$$P(X = k) = (1-p)^{k-1} p$$

Example: How many emails you must send before a customer makes a purchase. It helps predict the chance of success happening at each trial.

Continuous Data Distributions

A continuous distribution is used when the random variable can take any value within a specified range like when we analyze how much money a customer spends in a store then the amount can be any real number including decimals like 25.75,50.23, etc.

In continuous distributions the Probability Density Function (PDF) shows how the probabilities are spread across the possible values. The area under the curve of this PDF represents the probability of the random variable falling within a certain range.

1. Normal Distribution

The normal distribution, or bell curve, is one of the most common data distributions. Most values cluster around the mean, with fewer values farther away, forming a symmetrical shape. It's perfect for modeling things like people's heights.

- Mean is the center of the curve
- Symmetrical distribution (left and right sides mirror each other)
- Standard deviation shows how spread out the data is
- Smaller standard deviation means data is closer to the mean

PDF Formula:
$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Where:

- μ = mean
- σ = standard deviation

2. Exponential Distribution

The exponential distribution models the time between events happening independently and continuously. For example, the time between customer arrivals at a store.

PDF Formula:
$f(x) = \lambda e^{-\lambda x}, \quad x \ge 0$

Other Continuous Distributions (as listed)

- **Uniform Distribution**
  PDF: $f(x) = \frac{1}{b-a}, \quad a \le x \le b$
- **Beta Distribution**
  PDF: $f(x) = \frac{x^{\alpha-1}(1-x)^{\beta-1}}{B(\alpha,\beta)}$
- **Gamma Distribution**
  PDF: $f(x) = \frac{x^{k-1} e^{-x/\theta}}{\theta^k \Gamma(k)}, \quad x \ge 0$
- **Chi-Square Distribution**
  PDF: $f(x) = \frac{1}{2^{k/2}\Gamma(k/2)} x^{k/2-1} e^{-x/2}$
- **Log-Normal Distribution**
  PDF: $f(x) = \frac{1}{x\sigma\sqrt{2\pi}} \exp\left(-\frac{(\ln x - \mu)^2}{2\sigma^2}\right)$

Comparison of Probability Distributions (Table)

Distributions	Key Features	Usage
Normal Distributions	Used to adjust data to make it easier to analyze and to find unusual values like errors or outliers.	Used for feature scaling, model assumptions, and anomaly detection.
Exponential Distributions	Measures how long it takes for something to happen like waiting for an event.	Helps predict when a server might crash or how long it will take for customers to arrive at a store.
Uniform Distributions	Every possible outcome is equally likely; no outcome is more likely than another.	Used for picking random samples from a group.
Beta Distributions	Helps update guesses about chances based on new information.	Useful for A/B testing and figuring out how often people click on links.
Gamma Distributions	Measures the total time taken for several events to happen one after another.	Helps predict when systems might fail and assess risks.
Chi-Square Distributions	Checks if there is a relationship between different categories of data.	Helps analyze customer survey results to see if different groups have different opinions or behaviors.
Log-Normal Distributions	Shows how things grow over time especially when growth happens in steps.	Used for predicting stock prices and understanding income distribution.
Binomial Distributions	Models the number of successes in multiple trials.	Useful for determining the probability of a certain number of successes in a fixed number of trials.
Bernoulli Distributions	Models a single trial with two outcomes (success/failure).	Mostly used in quality control to assess pass/fail situations.

Poisson Distributions	Finds the number of events occurring in a fixed interval of time or space.	Helps predict the number of customer arrivals at a store during an hour.
Geometric Distributions	Helps find number of trials until the first success occurs.	Useful for understanding how many attempts it takes before achieving the first success (e.g., flipping a coin until heads).

Sampling Distributions in Data Science

What is Sampling distributions?

A sampling distribution is a statistical idea that helps us understand data better. It shows the values of a statistic when we take lots of samples from a population. For example, if we want to know the average height of people in a city, we might take many random groups and find their average height. The sampling distribution helps us understand the potential variability in average heights. By analyzing this distribution, entities like governments and businesses can make more informed decisions based on their collected data.

Importance Sampling Distribution in Data Science

Sampling distributions allow data scientists to:

- Estimate Population Parameters: By analyzing the distribution of sample statistics, data scientists can make inferences about population parameters (e.g., population mean or proportion).
- Quantify Uncertainty: Sampling distributions provide a measure of the variability of a statistic, which is crucial for constructing confidence intervals and hypothesis tests.
- Model Performance Evaluation: They help in understanding the variability and performance of models, especially when dealing with small datasets or conducting resampling techniques like bootstrap.

Types of Sampling distributions

1. Sampling Distribution of the Sample Mean (\bar{x})

If the population is normally distributed or the sample size is sufficiently large (according to the Central Limit Theorem), the sampling distribution of the sample mean is approximately normal with mean (\mu) and standard error (\frac{\sigma}{\sqrt{n}}).

2. Sampling Distribution of the Sample Proportion ((p))

If the conditions for using the normal approximation to the binomial distribution are met (e.g., large sample size, np ≥ 10, n(1-p) ≥ 10), the sampling distribution of the sample proportion is approximately normal with mean (p) and standard error \sqrt{\frac{p(1-p)}{n}}

3. Sampling Distribution of the Sample Variance ((S^2))

Population is normally distributed, the sampling distribution of the sample variance follows a chi-square distribution with (n-1) degrees of freedom

Central Limit Theorem in Sampling Distributions

The Central Limit Theorem (CLT) is a fundamental concept in statistics. It states that when we take many samples from any population and calculate their averages, the distribution of these sample averages will tend to follow a bell-shaped

curve, or a normal distribution , regardless of the shape of the original population distribution.

This is important because it allows us to make predictions and draw conclusions about population parameters, like the population mean, even when we only have information about sample averages. So basically, the CLT helps us understand and predict the behavior of sample averages, making it a key tool in statistical analysis.

How CLT Shapes Sampling Distributions?

The Central Limit Theorem (CLT) shapes sampling distributions by providing insights into how the distribution of sample means behaves as the sample size increases.

- Approach to Normality: Regardless of the shape of the population distribution, the sampling distribution of the sample mean tends to become more normally distributed as the sample size increases. This means that even if the population distribution is not normal, the distribution of sample means will approximate a normal distribution.
- Stability of Parameters: The mean of the sampling distribution of the sample mean is equal to the population mean. This means that as you take larger and larger samples, the average of those samples will converge to the population mean.
- Decrease in Variability: The variance of the sampling distribution of the sample mean decreases as the sample size increases. Specifically, the variance of the sample mean is equal to the population variance divided by the sample size. This means that larger sample sizes lead to less variability in the sample means.
- Predictability: With larger sample sizes, the sampling distribution becomes more predictable and consistent. This is because the sample mean tends to be closer to the population mean, and the spread of the distribution becomes narrower.
- Use of Normal Distribution: The CLT allows statisticians to use the properties of the normal distribution to make inferences about population parameters. For example, confidence intervals and hypothesis tests often rely on the assumption of normality, which is justified by the CLT when dealing with sample means.

Understanding the Distribution of Sample Means

- The Distribution of Sample Means, also known as the sampling distribution of the sample mean, depicts the distribution of sample means obtained from multiple samples of the same size taken from a population.
- Central Limit Theorem states that as the sample size increases, distribution of sample means approaches a normal distribution, regardless of the shape of the population distribution.
- This distribution is crucial because it allows us to make inferences about population parameters based on sample statistics, such as estimating population means and constructing confidence intervals.

```
 import numpy as np
import matplotlib.pyplot as plt

# Generate population data (normal distribution)
population_mean = 50
population_std = 10
population_size = 1000
population_data = np.random.normal(population_mean, population_std, population_size)

# Sample size and number of samples
sample_size = 30
num_samples = 1000

# Calculate sample means
sample_means = [np.mean(np.random.choice(population_data, sample_size)) for _ in range(num_samples)]
```

```
# Plot histogram of sample means
plt.hist(sample_means, bins=30, edgecolor='black')
plt.xlabel('Sample Mean')
plt.ylabel('Frequency')
plt.title('Distribution of Sample Means')
plt.show()
```

Output:

Figure_1 – Distribution of Sample Means

The output graph is a histogram that peaks around the population of 50 showing that the average of the sample means is a good estimator of the population mean.

Understanding the Distribution of Sample Proportions

- The Distribution of Sample Proportions describes the distribution of sample proportions (e.g., the proportion of successes) obtained from multiple samples of the same size taken from a population.
- When the sample size is sufficiently large (usually greater than 30) and sampling method is random, the distribution of sample proportions can be approximated by a normal distribution.
- The distribution is also fundamental in various applications, such as estimating population proportions, testing hypotheses about proportions, and constructing confidence intervals for proportions.

```
 import numpy as np
import matplotlib.pyplot as plt

# Generate population data (binomial distribution)
population_size = 1000
population_proportion = 0.6
population_data = np.random.binomial(1, population_proportion, population_size)

# Sample size and number of samples
sample_size = 50
num_samples = 1000

# Calculate sample proportions
sample_proportions = [np.mean(np.random.choice(population_data, sample_size, replace=True)) for _ in ran

# Plot histogram of sample proportions
plt.hist(sample_proportions, bins=30, edgecolor='black')
plt.xlabel('Sample Proportion')
plt.ylabel('Frequency')
plt.title('Distribution of Sample Proportions')
plt.show()
```

Output:

Figure_1 – Distribution of Sample Proportions

The output graph displays the frequency distribution of the generate data, the peak of the histogram the is around population proportion (0.6), indicating that the average of the sample proportions is a good estimator of the population proportion.

Understanding the Distribution of Sample Variances

- The Distribution of Sample Variances showcases the distribution of sample variances obtained from multiple samples of the same size taken from a population.
- Unlike the sample mean, distribution of sample variances does not necessarily follow a normal distribution, especially for small sample sizes or non-normally distributed populations.
- Understanding this distribution is essential in statistical analysis, particularly in assessing the variability of data and making inferences about population variances.

```
 import numpy as np
import matplotlib.pyplot as plt

# Generate population data (uniform distribution)
population_data = np.random.uniform(0, 100, 1000)

# Sample size and number of samples
sample_size = 50
num_samples = 1000

# Calculate sample variances
sample_variances = [np.var(np.random.choice(population_data, sample_size)) for _ in range(num_samples)]

# Plot histogram of sample variances
plt.hist(sample_variances, bins=30, edgecolor='black')
plt.xlabel('Sample Variance')
plt.ylabel('Frequency')
plt.title('Distribution of Sample Variances')
plt.show()
```

Student's t-distribution in Statistics

The Student's t-distribution (or simply the t-distribution) is a probability distribution used in statistics when making inferences about a population mean, particularly when the sample size is small ($n \leq 30$) or the population standard deviation (σ) is unknown. It resembles the standard normal distribution but has heavier tails, allowing it to better handle variability in small samples. The t-score indicates how many estimated standard errors the sample mean ($\bar{x}$) is away from the population mean (μ).

Formula for the t-Score

The t-score (or t-statistic) quantifies how many estimated standard errors the sample mean ($\bar{x}$) is from the population mean:

$t = \frac{\bar{x} - \mu}{s \sqrt{n}}$

where,

- t = t-score,
- \bar{x} = sample mean
- μ = population mean,
- s = standard deviation of the sample,
- n = sample size

The t-score helps determine how far the sample mean is from the population mean under the assumption of random sampling.

When to Use the t-Distribution

Student's t Distribution is used when :

- The sample size is 30 or less than 30.
- The population standard deviation(σ) is unknown.
- The population distribution must be unimodal and skewed.

Interpretation of t-Distribution with Example

Suppose a researcher wants to estimate the average daily study time of students before exams. A random sample of 20 students reports an average (\bar{x}) of 4 hours with a sample standard deviation (s) of 1.5 hours. We want to construct a 90% confidence interval for the true population mean.

Given:

- \bar{x} = 4 hours, s = 1.5 hours, n = 20 and a 90% confidence level.
- Degrees of freedom = n – 1 = 19
- Critical t-value (from t-table) ≈ 1.729

CI = \bar{x} \pm t \times \frac{s}{\sqrt{n}}

Substituting the given values:

CI = 4 \pm 1.729 \times \frac{1.5}{\sqrt{20}} = (3.42, , 4.58)

Interpretation: We are 90% confident that the true average study time for all students lies between 3.42 and 4.58 hours per day. This range indicates where the actual population mean is most likely to fall, given the data and confidence level.

Implementation

Let's implement the example in Python using scipy.stats:

```
 import numpy as np
from scipy import stats

x_bar = 4
s = 1.5
n = 20
confidence = 0.90
df = n - 1
t_critical = stats.t.ppf((1 + confidence) / 2, df)
margin_of_error = t_critical * (s / np.sqrt(n))
lower_bound = x_bar - margin_of_error
```

```
upper_bound = x_bar + margin_of_error

print(f"t-critical value: {t_critical:.3f}")
print(f"Confidence Interval (90%): ({lower_bound:.2f}, {upper_bound:.2f})")
```

Output:

t-critical value: 1.729

Confidence Interval (90%): (3.42, 4.58)

Properties of the t-Distribution

- It is symmetric and bell-shaped, like the normal distribution.
- The variable t ranges from $-\infty$ to $+\infty$.
- As degrees of freedom (df) increase, the t-distribution approaches the standard normal distribution (Z).
- The mean, median and mode are all zero (for df > 1).
- Variance = df / (df − 2) for df > 2, indicating that as df increases, variance decreases.
- It has heavier tails than the normal distribution, making it more flexible for small samples.

Key Elements in Student's t-Distribution

1. t-Distribution Table

- Provides critical t-values for various confidence levels and degrees of freedom.
- Used to determine whether the calculated t-score falls in the rejection region of a hypothesis test.
- Example : At a 5% significance level ($\alpha = 0.05$), if the calculated $|t|$ exceeds the tabulated t-value, the difference between sample and population means is considered statistically significant.

2. t-Score

- t-Score measures how far a sample mean deviates from the population mean in terms of standard errors.
- Helps determine statistical significance and construct confidence intervals.
- Larger $|t|$ values indicate a greater difference between sample and population means.

3. p-Value

- p-Value represents the probability of obtaining a test result at least as extreme as the one observed, assuming the null hypothesis is true.
- A small p-value (typically < 0.05) suggests strong evidence against the null hypothesis, leading to its rejection.
- It can be obtained from statistical software or a t-table using the calculated t-score and degrees of freedom.

Applications

- Testing a Population Mean : Determine if a sample mean significantly differs from a known or hypothesized population mean.
- Comparing Two Means : Evaluate whether two independent or paired samples have different means.
- Testing Correlation : Assess if the correlation coefficient between two variables significantly differs from zero.

Limitations

- Assumes Normality : The t-distribution relies on the assumption that the underlying population is normally distributed. Significant deviations can lead to inaccurate results.
- Less Useful for Large Samples : As the sample size increases, the t-distribution approaches the normal distribution, making the latter more suitable for large datasets.
- Sensitive to Outliers : Extreme values can distort results, especially in small samples where the t-distribution's heavy tails amplify their effect.
- Requires Random Sampling : Valid conclusions depend on random, independent observations. Violating these assumptions reduces the reliability of inferences.

Difference Between T-Distribution and Normal Distribution

Aspect	t-Distribution	Normal Distribution
Definition	Defined by degrees of freedom (df) depending on sample size	Defined by mean (μ) and standard deviation (σ)
Sample Size	Used for small samples ($n \leq 30$)	Used for large samples ($n > 30$)
Standard Deviation	Unknown (estimated from sample)	Known
Shape	Heavier tails more prone to extreme values	Lighter tails, data closer to mean
Application	Hypothesis testing when σ is unknown	When σ is known or sample size is large
Range of Critical Values	Wider range due to more uncertainty	Narrower range with less uncertainty